4. Vaccum Cleaner Agent :

```
VACCUM-AGENT (location, status_A, status_B)
    if location = A and status_A = DIRTY then
        return SUCK
    else if location = B and status_B = DIRTY then
        return SUCK
    else if status_A = DIRTY then
        return LEFT
    else if status_B = DIRTY then
        return RIGHT
    else
        return NO_OP

VACCUM-STIMILATION (location, status_A, status_B)
    while TRUE do
        action ← VACCUM-AGENT (location, status_A,
                               status_B)
        if action = NO_OP then
            stop
        if action = SUCK and location = A then
            status_A ← Clean
        else if action = SUCK and location = B then
            status_B ← clean
        else if action = LEFT then
            location ← A
        else if action = RIGHT then
            location ← B
        display action and current environment
                                        State
    display final location, status_A, status_B
```

Test cases :-

1. Both Rooms Dirty, Starting in A

Input : Initial location : A
Room A: Dirty
Room B : Dirty

Output:

Initial State → Location : A, Room A : Dirty, Room B : Dirty
Step 1 : Action = SUCK | New state → location : A, Room A : Clean, Room B : Dirty
Step 2 : Action = RIGHT | New state → Location : B, Room A : Clean, Room B : Dirty
Step 3 : Action = SUCK | New state → Location : B, Room A : Clean, Room B : Clean

Step 4 : Action = NO-OP

Final State → location : B, Room A : Clean, Room B : Clean

2. First room clean, second room dirty, Start A

Input : Initial location : A
Room A: Clean
Room B : Dirty

Output:

Steps	Action	State location	Room A	Room B
Initial	-	A	Clean	Dirty
1	RIGHT	B	Clean	Dirty
2	SUCK	B	Clean	Clean
3	NO-OP	B	Clean	Clean
Final	-	B	clean	clean

3. Both rooms already clean, start B

Input : Initial location : B
Room A : Clean
Room B : Clean

Output:

Steps	Action	State: location	State: Room A	State: Room B
Initial	-	B	Clean	Clean
1	NO-OP	B	Clean	Clean
Final	-	B	clean	clean

4. Both Rooms dirty, Start B

Input: Inital location: B
Room A: dirty
Room B: dirty

Output:

Steps	Action	State: location	State: Room A	State: Room B
Initial	-	B	dirty	dirty
1	SUCK	B	dirty	clean
2	LEFT	A	dirty	clean
3	SUCK	A	clean	clean
4 ~~Final~~	NO-OP	A	Clean	clean

5. Room A dirty, B clean, Start B

Initial	-	B	dirty	clean
1	LEFT	A	dirty	clean
2	SUCK	A	clean	clean
3	NO-OP	A	clean	clean

6. Room A clean, B dirty, Start B

Initial	-	B	~~d~~clean	dirty
1	SUCK	B	clean	clean
2	LEFT	A	clean	clean
3	NO-OP	A	clean	clean

7. Both rooms clean, Start A

Initial	-	A	clean	clean
1	NO-OP	A	clean	clean